The pointed Lipschitz space need not be complete

A negative answer to Remark 3.3 of arXiv:2410.10677

Run `fa_banach_001` — `agent_lane_03`

13 August 2026

Status. High-confidence candidate full counterexample, pending human review. The proposed completeness fails already for maps $\mathbb{R} \to \mathbb{R}$.

The source question

Let M, N be metric spaces with distinguished point 0. Write $\mathrm{Lip}_0(M, N)$ for the Lipschitz maps $f : M \to N$ with $f(0) = 0$, and

$$d_e(f, g) = \sup_{x \neq 0} \frac{d(f(x), g(x))}{d(x, 0)}.$$

Karn and Mandal prove that the larger space $E(M, N)$ of extensively bounded maps is complete whenever N is complete. Their Remark 3.3 asks whether $\mathrm{Lip}_0(M, N)$ is also complete under the restricted metric d_e.

A generalization of Lipschitz mappings 11

Thus the metric d_e induces a norm $\|\cdot\|_e$ on $E(M, Y)$ given by

$$\|f\|_e = d_e(f, 0)$$

for all $f \in E(M, Y)$ whenever Y is a (real) normed linear space.

In particular, if Y is a Banach space, then $(E(M, Y), \|\cdot\|_e)$ is also a Banach space. $\square$

Remark 3.3. In general, there is no specific metric defined on $Lip_0(M, N)$. However, being a subspace of $E(M, N)$, we can provide the restriction of d_e on it. But we do not know whether, in general, $Lip_0(M, N)$ is complete with this metric or not, even assuming the completeness of N.

Example 3.4. *In general, $E(M, N)$ and $C(M, N)$ are distinct spaces. For example, let $M = N = \mathbb{R}$ and consider the functions*

The exact source page is bundled as `source_excerpt_remark_3_3_page_11.pdf`.

Theorem 1. *With the usual metric and distinguished point zero,* $\mathrm{Lip}_0(\mathbb{R}, \mathbb{R})$ *is not complete under d_e. More precisely, there are compactly supported Lipschitz functions $f_n : \mathbb{R} \to \mathbb{R}$ that are d_e-Cauchy and converge in $E(\mathbb{R}, \mathbb{R})$ to a continuous extensively bounded function that is not Lipschitz.*

Construction and proof

Put $a = \frac{1}{2}$ and

$$\eta(x) = \big(1 - 4|x-a|\big)_+, \qquad r_n(x) = \big((x-a)^2 + n^{-2}\big)^{1/4}, \qquad r(x) = |x-a|^{1/2}.$$

Define

$$f_n(x) = \eta(x) r_n(x), \qquad f(x) = \eta(x) r(x).$$

The tent function η is 4-Lipschitz and is supported in $[\frac{1}{4}, \frac{3}{4}]$. For each fixed n, r_n is continuously differentiable and

$$|r_n'(x)| = \frac{|x-a|}{2((x-a)^2 + n^{-2})^{3/4}} \le \frac{\sqrt{n}}{2}.$$

Thus f_n is a compactly supported Lipschitz function. Since its support misses zero, $f_n(0) = 0$, so $f_n \in \mathrm{Lip}_0(\mathbb{R}, \mathbb{R})$.

For $u, v \ge 0$ and $0 < \alpha < 1$, subadditivity gives $(u+v)^\alpha \le u^\alpha + v^\alpha$. Taking $u = (x-a)^2$, $v = n^{-2}$, and $\alpha = \frac{1}{4}$ yields

$$0 \le r_n(x) - r(x) \le n^{-1/2}.$$

Outside $[\frac{1}{4}, \frac{3}{4}]$ both f_n and f vanish, while on this interval $|x| \ge \frac{1}{4}$. Consequently

$$d_e(f_n, f) = \sup_{x \neq 0} \frac{|f_n(x) - f(x)|}{|x|} \le \frac{4}{\sqrt{n}}. \tag{1}$$

In particular, $d_e(f_n, f_m) \le 4(n^{-1/2} + m^{-1/2})$, so (f_n) is d_e-Cauchy.

The limit f is continuous, compactly supported, and extensively bounded: on its support $|x| \ge \frac{1}{4}$ and $|f(x)| \le \frac{1}{2}$, hence $|f(x)|/|x| \le 2$. It is not Lipschitz. Indeed, for $0 < h < \frac{1}{4}$,

$$f(a) = 0, \qquad f(a+h) = (1-4h)\sqrt{h}, \qquad \frac{|f(a+h) - f(a)|}{h} = \frac{1-4h}{\sqrt{h}} \longrightarrow \infty.$$

Therefore the d_e-limit in the complete metric space $E(\mathbb{R}, \mathbb{R})$ does not belong to $\mathrm{Lip}_0(\mathbb{R}, \mathbb{R})$. Uniqueness of limits rules out any other Lip_0 limit, proving Theorem 1. □

Corollary 1. *For every nonzero Banach space Y, the space $\mathrm{Lip}_0(\mathbb{R}, Y)$ is incomplete under d_e.*

Proof. Fix $y \in Y$ with $\|y\| = 1$ and replace f_n, f by $f_n y, fy$. All estimates and the cusp obstruction are unchanged. □

Proof Intuition

The metric d_e controls the size of a map only relative to the single distinguished point. On a region bounded away from zero, it is merely a weighted uniform metric. Uniform convergence preserves continuity but not a uniform Lipschitz constant. We therefore regularize a square-root cusp inside such a region. The Lipschitz constants of the approximants diverge, but d_e sees only their uniformly vanishing vertical error.

This also shows that the failure occurs inside the source paper's complete continuous subspace $E_c(\mathbb{R}, \mathbb{R})$: continuity survives, while global Lipschitz regularity does not.

Scope, novelty audit, and review recommendation

The theorem fully answers the universal question negatively, with both domain and codomain Banach spaces. It does not characterize the pointed metric spaces for which completeness does hold.

Exact-question, exact-title, author, arXiv-id, $\mathrm{Lip}_0(M, N)$, d_e, and “extensively bounded” searches through 13 August 2026 found the source and a current index still listing the question as open, but no later explicit answer or equivalent counterexample. This is a bounded novelty audit, not a priority claim. Human review should check the definition match, estimate (1), and the cusp quotient; no external theorem is used.

Reference

[1] A. K. Karn and A. Mandal, *A generalization of Lipschitz mappings*, arXiv:2410.10677, Remark 3.3, PDF p. 11.